C6.1 What useful products can be made from acids?

Teaching and learning narrative	Assessable learning outcomes *Learners will be required to:*		*Linked learning opportunities*
Many products that we use every day are based on the chemistry of acid reactions. Products made using acids include cleaning products, pharmaceutical products and food additives. In addition, acids are made on an industrial scale to be used to make bulk chemicals such as fertilisers. Acids react in neutralisation reactions with metals, hydroxides and carbonates. All neutralisation reactions produce salts, which have a wide range of uses and can be made on an industrial scale. **The strength of an acid depends on the degree of ionisation and hence the concentration of H^+ ions, which determines the reactivity of the acid. The pH of a solution is a measure of the concentration of H^+ ions in the solution. Strong acids ionise completely in solution, weak acids do not. Both strong and weak acids can be prepared at a range of different concentrations (i.e. different amounts of substance per unit volume).** **Weak acids and strong acids of the same concentration have different pH values. Weak acids are less reactive than strong acids of the same concentration (for example they react more slowly with metals and carbonates).**	1.	recall that acids react with some metals and with carbonates and write equations predicting products from given reactants	**Specification links** • Writing formulae, balanced symbol and ionic equations.(C3.2) • Concentration of solutions (C5.4) **Practical Work:** • Reactions of acids and preparation of salts. • pH testing • Investigating strong and weak acid reactivity. • use of indicators to test strong and weak acids, making standard solutions using volumetric flasks.
	2.	describe practical procedures to make salts to include appropriate use of filtration, evaporation, crystallisation and drying *PAG7*	
	3.	use the formulae of common ions to deduce the formula of a compound	
	4.	recall that relative acidity and alkalinity are measured by pH including the use of universal indicator and pH meters	
	5.	**use and explain the terms dilute and concentrated (amount of substance) and weak and strong (degree of ionisation) in relation to acids including differences in reactivity with metals and carbonates**	
	6.	**use the idea that as hydrogen ion concentration increases by a factor of ten the pH value of a solution decreases by one**	
	7.	**describe neutrality and relative acidity and alkalinity in terms of the effect of the concentration of hydrogen ions on the numerical value of pH (whole numbers only)**	